NovaTech Industries — Data Policy Approval & Synthetic Dataset Design

Checkpoint: 2.5

Document ID: NT-DATA-025

Version: 1.0

Status: Proposed for business review — not approved

Date: 21 October 2026

Related requirements: FR-005, BR-001–BR-006, DV-001–DV-010

Classification: Internal training case study; company and data are fictional

1. Purpose and decision context

This document defines proposed operational policies and the synthetic-data plan for the NovaTech Industries Supplier Recommendation Engine. It translates the Business Requirements Document, Supplier Evaluation Requirements, Score Normalization specification, and Supplier Evaluation Data Dictionary into testable data-generation rules.

Previously agreed: Eligible quotations are ranked using a weighted score: total purchasing cost 40%, delivery lead time 25%, product quality 20%, and supplier reliability 15%. Mandatory eligibility rules are evaluated before scoring. The Procurement Officer reviews the recommendation before an order is issued.

Not yet approved: Rules for insufficient historical performance, comparison currencies, minimum observation counts, same-day delivery, score ties, and approval thresholds. This document proposes defaults; none should be treated as approved business policy until confirmed.

2. Policy decisions requiring approval

ID	Policy area	Proposed policy	Rationale	Status
DP-001	Evaluation currency	Compare all quotations in EUR; retain original currency, source, rate, and effective date	Reproducible cross-border comparisons	Proposed
DP-002	Total cost	Unit price × requested quantity + attributable freight + duties + nonrecoverable fees; avoid double-counting	Comparable landed-cost proxy	Proposed
DP-003	Historical window	Use the 12 months preceding evaluation date	Consistent recent performance	Proposed
DP-004	Quality sample	Require at least 100 inspected units for automated quality scoring	Avoid unstable rates from tiny samples	Proposed
DP-005	Reliability sample	Require at least 10 completed deliveries for automated reliability scoring	Avoid unstable rates from tiny samples	Proposed
DP-006	Insufficient history	Mark as REVIEW_REQUIRED; do not fabricate scores or automatically recommend	Prevent unsupported comparisons	Proposed

ID	Policy area	Proposed policy	Rationale	Status
DP-007	Same-day lead time	Accept zero days, award delivery score 100 if shortest; otherwise use positive-day ratio	Avoid division by zero	Proposed
DP-008	Equal scores	Tie-break by lower total cost, then shorter lead time, then stable quotation ID	Deterministic output	Proposed
DP-009	Weight configuration	Weights must be nonnegative and total exactly 100%; save version with evaluation	Auditability	Proposed
DP-010	Human approval	A recommendation never creates or sends a purchase order without approval	Maintain control over spending	Proposed
DP-011	Missing required quotation fields	Exclude incomplete quotation from automated ranking and record reason	Data integrity	Proposed
DP-012	Missing exchange rate	Do not score quotation until a valid rate is available	Prevent misleading cost rankings	Proposed

2.1 Scoring definitions

For each eligible, sufficiently documented quotation:

• cost_score = 100 × minimum_eligible_total_cost / quotation_total_cost (strictly positive cost).

• delivery_score = 100 for a zero-day offer when it is the minimum; otherwise 100 × minimum_eligible_lead_time / quotation_lead_time when all compared lead times are positive. If the minimum is zero, zero-day offers receive 100 and positive-day offers receive 0; this edge-case rule is **proposed** and subject to review.

• quality_score = 100 × accepted_units / inspected_units, provided the quality sample threshold is met.

• reliability_score = 100 × on_time_deliveries / completed_deliveries, provided the reliability sample threshold is met.

• weighted_score = 0.40 × cost_score + 0.25 × delivery_score + 0.20 × quality_score + 0.15 × reliability_score.

Important: The relative cost and delivery scores depend on the comparison set. Save the set of eligible quotation IDs, raw values, exchange rates, criteria weights, and calculation version with every completed evaluation.

3. Initial synthetic dataset scope

The first release deliberately uses a small dataset that is easy to inspect manually and run on an 8 GB Intel Mac. It is a subset of the larger enterprise simulation.

Dataset	Initial target	Grain (one row represents)	Key fields
suppliers.csv	12	One supplier	supplier_id, name, country, status, approved_categories
products.csv	20	One product	product_id, sku, category, unit_of_measure
supplier_products.csv	45	One supplier-product offering	supplier_id, product_id, min_order_qty, available_qty
purchase_requisitions.csv	15	One requisition header	requisition_id, request_date, required_date, status

Dataset	Initial target	Grain (one row represents)	Key fields
requisition_items.csv	25	One requested product line	requisition_item_id, requisition_id, product_id, quantity
quotations.csv	35	One supplier quotation header	quotation_id, supplier_id, issued_date, valid_until, currency
quotation_items.csv	55	One quotation product line	quotation_item_id, quotation_id, product_id, quantity, unit_price, lead_time_days
supplier_performance.csv	12	One supplier's 12-month metrics	supplier_id, inspected_units, accepted_units, completed_deliveries, on_time_deliveries
exchange_rates.csv	8	One dated currency conversion	currency, rate_to_eur, effective_date, source
evaluation_criteria.csv	4	One criterion and weight	criterion, weight_pct, policy_version

Initial target total: 231 records across 10 CSV files. Targets are provisional and may change to preserve referential integrity. The larger 12-table evaluation schema and wider procurement/logistics datasets remain future phases.

4. Data-generation principles

1. **Deterministic:** Use a fixed random seed (proposed: 42) so test datasets can be reproduced.

2. **Fictional:** Generate synthetic suppliers, products, quotations, and transactions. Do not use real supplier personal data or confidential business records.

3. **Referentially valid:** Every foreign key must refer to an existing parent record; no orphan quotation items.

4. **Chronologically valid:** Request dates precede required dates; quotation issue dates precede expiry; evaluations use rates and performance available as of evaluation time.

5. **Commercially plausible:** Prices, minimum quantities, lead times, and available quantities vary by product category and supplier region.

6. **Consistent units:** Quantities use each product's unit of measure; lead times use calendar days; money uses explicit currency codes.

7. **Separated data quality:** Keep the clean reference dataset separate from deliberately corrupted test fixtures.

8. **Documented assumptions:** Record generator version, seed, row counts, scenario flags, and any simplifications in a manifest.

5. Required test scenarios

Scenario ID	Business scenario	Expected outcome
SC-001	Multiple eligible quotations	Rank by weighted score
SC-002	Cheapest supplier has slower delivery	Weighted score may favor another supplier
SC-003	Supplier inactive	Exclude and record reason
SC-004	Quotation expired	Exclude and record reason
SC-005	Quantity unavailable or below minimum	Exclude and record reason
SC-006	Delivery misses required date	Exclude and record reason
SC-007	Supplier lacks sufficient quality history	Require manual review

Scenario ID	Business scenario	Expected outcome
SC-008	Supplier lacks sufficient delivery history	Require manual review
SC-009	Two quotations tie on weighted score	Apply deterministic tie-break
SC-010	Quotation in USD with recorded EUR rate	Convert and retain conversion evidence
SC-011	Exchange rate unavailable	Block automated scoring
SC-012	No eligible quotations	Return clear no-recommendation outcome
SC-013	Invalid negative price or lead time	Reject during validation
SC-014	Same-day delivery	Apply approved zero-day scoring rule

6. Dataset storage conventions

```text
ai-procurement-platform/
■■■ docs/
■ ■■■ data/
■ ■■■ supplier-evaluation-data-dictionary.md
■ ■■■ data-policy-and-synthetic-dataset-design.md
■■■ data/
■ ■■■ synthetic/
■ ■ ■■■ v1/
■ ■ ■■■ suppliers.csv
■ ■ ■■■ products.csv
■ ■ ■■■ ...
■ ■■■ fixtures/
■ ■■■ invalid/
■■■ src/
■■■ data_generation/
```

Conventions: UTF-8 encoded CSV; ISO dates (YYYY-MM-DD); ISO 4217 currency codes; dot decimal separator; stable string identifiers (SUP001, PRD001, QUO001); snake_case field names; blank values only where explicitly allowed. Avoid storing generated secrets or personally identifying information in Git.

7. Quality gates before coding FR-005

• All IDs unique within their entity and all foreign keys resolve.

• No duplicate supplier-product pairs in the initial catalog.

• Valid date sequences and quotation status/expiry rules.

• Price, quantity, lead time, and exchange-rate validations pass.

• Quality and reliability numerators never exceed denominators.

• Weight percentages total exactly 100.

• All 14 test scenarios are represented and have expected results.

• Running the generator twice with the same seed produces identical outputs.

• A data manifest records version, seed, assumptions, and file row counts.

8. Decisions and sign-off

Recommended starting policy: Approve DP-001 through DP-012 as a *development baseline*, with particular review of DP-004/DP-005 (sample sizes), DP-006 (new suppliers), and DP-007 (same-day scoring). These are design proposals, not facts about an actual company.

Business owner: Procurement Manager (role in simulated organization)

Technical owner: Business Transformation & Analytics Team

Approval status: Pending user confirmation

9. Next checkpoint

Checkpoint 2.6 — Synthetic Data Generator, Part 1: After policy approval, create the directory structure and a small reproducible Python generator for suppliers.csv, products.csv, and supplier_products.csv. Validate keys and inspect outputs before adding quotations and evaluations.